

‘Can a machine do this job?’ is the wrong question

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Whenever AI automation anxiety arises, optimists like to point to the bank teller. US vice-president JD Vance repeated the idea last year: ATMs automated the teller’s core task and yet teller employment rose for three decades. The implicit assumption is that if automation could not kill the teller, surely AI will not kill the accountant.

As economics writer David Oks has noted, this reassurance is premature. Once customers could deposit a cheque by photographing it and send money without visiting a branch, the teller was replaced — not by a better machine in the bank but by the customer outside it.

The distinction matters, and it turns on a mechanism economists have understood since the 19th century. When a technology automates tasks inside an existing service, it can trigger a Jevons paradox: the service becomes cheaper, demand expands, and employment grows. That is what ATMs did and it is the reason automation has so often failed to produce mass unemployment.

But the paradox has a condition: it works only when the technology makes the existing service model more efficient. When a technology lets people do the work themselves, demand for the service collapses.

The sociologist Jonathan Gershuny identified this pattern in 1978. Modern economies, he argued, were not heading towards a service utopia but a self-service economy in which households would absorb the work themselves. The washing machine illustrates this: it did not automate the laundress’s job — it gave customers the means to do without her.

The pattern has been repeating ever since. Self-checkout handed scanning and bagging to the shopper. The internet gave travellers direct access to flight schedules and hotel reviews that agents once controlled. Online brokerages put a trading terminal in every pocket.

AI extends this mechanism even to the manual trades, the supposed safe haven of the AI age. If a homeowner can ask a chatbot why their boiler is losing pressure, heating engineers may lose call-outs. Nor are professions immune: doctors may find patients have decoded test results before they arrive.

This solves a problem for companies in the process. As Christian Catalini of the MIT Cryptoeconomics Lab has argued, when AI pushes the cost of execution towards zero, the binding constraint becomes human verification — our capacity to validate outcomes and take responsibility. Self-service offloads that burden on to the customer.

This shift has broader macroeconomic implications. When work shifts to the consumer, it vanishes from the economy that statisticians measure. A company that replaces a billing department with a chatbot interface records lower costs and higher output per worker. The national accounts register a productivity gain. But the hours that patients spend decoding their own tests appear nowhere — not in labour statistics, not in GDP. As AI self-service expands into professional domains, this blind spot will grow.

Policymakers who rely on those indicators to judge whether AI is delivering benefits may be missing a deeper shift. The great achievement of modern capitalism was to move activity from the household into the market — converting domestic production into paid specialisation, creating jobs and making output visible to the national accounts. AI-enabled self-service is quietly reversing that centuries-long trend.

The automation question — can a machine do this job? — would never have predicted the laundress's decline. No robot could walk to the well and handwash linens. But the washing machine did not need to. The self-service question — can the customer do without this job? — would have predicted it. If we keep asking the first question about AI, we will keep looking in the wrong place.